

Mingtai Zhang

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Education

Guangxi University | School of Resources, Environment and Materials

2022 – 2026

B.Eng. in Materials Science and Engineering

Main courses were physical chemistry, fundamentals of materials science, physical metallurgy, principles of metallurgical processes, materials characterization, electrical engineering, mechanical design.

I taught myself mathematics and programming. I did the computational work below with a research group at Northeastern University (Qinhuangdao). I ran a full experimental thesis in electrochemistry over the same period.

Research Summary

I have worked on five projects in four fields. In each one I built the part that tests whether a claim is true. In my first-author work I defined what a companion agent may write about a real user. I showed that training can improve the replies of a model. The same training corrupts the user profile that the model writes. A test of replies alone never sees this failure. In coherent imaging I built the test harness for a result on how many measurements are needed. In test-time selection I ran the whole experiment that verifies a lower bound on the labels needed to measure accuracy. The habit started in electrochemistry. Every claim about the interface had to hold in several independent measurements. I want to work on measurement design for systems that cannot be observed directly. My question is what a test must cover, at what cost, before a reported improvement can be believed.

Research Interests

Evaluation and auditing of LLM systems | memory, attribution, trustworthy human–AI interaction | LLM agents that reason from evidence | inverse problems, sample complexity | AI for experimental science

Publications and Manuscripts

Each venue has an anonymity policy. Co-author lists for papers under double-blind review stay withheld until notification.

Under Review

- [1] Paper on attribution boundaries in conversational AI memory (title withheld during review) **First author**; under double-blind review, a top-tier HCI conference.
- [2] Paper on the accuracy–evaluation trade-off in test-time selection (title withheld during review) **Second author**; under double-blind review, a top-tier machine-learning conference.
- [3] Yuchen Miao, **Mingtai Zhang**, Zijun Wang, Ruobing Han, Chang Han, Siyang Xu. “Two-Scan Recovery of Sparse Defects in Ptychography.” **Second author**; under review, *IEEE Signal Processing Letters*, 2026. (My CRediT roles were data curation and software.)

Accepted

- [4] Yuchen Miao, Zijun Wang, Chang Han, Yurui Shi, **Mingtai Zhang**, Siyang Xu. “The ‘10th Juror’: Open-Set Standpoint Screening for Bureaucratic Bias Detection.” *EMNLP 2026* (Main Conference) [CORE A*; CCF-B].
- [5] Guang Zhang, Dong Yang, Kewang Yang, Farwa Mushtaq, Meng Zhang, **Mingtai Zhang**, Jieshang Lu, Yucheng Wen, Meinan Liu. “Modified Ether Electrolyte by Siloxane Additive with Abundant Si–O Bonds for High-Voltage Lithium Metal Batteries.” *Journal of Physics: Conference Series* **3165**, 012009 (2025); CEAMS 2025 [EI-indexed]. DOI 10.1088/1742-6596/3165/1/012009.

Research Experience

Attribution and Memory in Conversational AI Agents

2026

First author, under double-blind review | problem, measures, experiments, writing

- **Problem.** A user plays a fictional character in a chat. The companion agent can store what the character says as a fact about the real person. I wrote one rule for this case. A statement made only inside a story must not appear later as a fact about the real user with no note on its source.
- **Two channels.** I scored the reply channel apart from the stored memory channel. I did not score overall response quality. For replies I measured how often the model treats a story fact as true of the user. For written profiles I measured how often the stored profile records a story fact as true. Each measure had written scoring rules. I checked how well the rules covered the data. A second measure on how the model positions itself toward the user stayed clearly secondary.
- **Ground truth.** I built a controlled corpus of rule-generated Chinese episodes. Each episode holds one value for the character, one value for the real user, on the same attribute. Every error can be traced to one source. I scored tens of thousands of replies. I scored a large set of written profiles in a multi-condition design over boundary cues, model size, thinking settings, writer prompts.
- **Main finding.** Two fine-tuned variants improved attribution in replies. Neither improved profile writing in a consistent way. Fine-tuning aimed at the conversation raised profile corruption in every reported condition summary. A test of replies alone would have scored both variants as successes.

Accuracy and Evaluation Cost in Test-Time Selection

2026

Second author, under double-blind review | designed and ran the whole experimental verification

- A system can generate several candidate answers. It keeps the best one. This raises accuracy. It makes accuracy harder to measure. The kept answers are rare in the labels already collected. The paper gives the best accuracy reachable for a given number of candidates when the check reuses labels from the base

model. It proves a worst-case error for every rule of this kind. The error grows with the number of candidates per available label. Fixed precision costs more labels as the candidate budget grows.

- Two limits shape the result. One limit is which output distributions a candidate pool can reach. The other limit is how hard each distribution is to audit. The two limits define a frontier between accuracy and audit cost. The frontier gives less weight to correctness in regions with thin label coverage. I tested frontier policies with cross-fitting on four frozen public reasoning banks. They beat uniform best-of-N mixtures at equal audit budget. They matched two fully tuned baselines.

Two-Scan Recovery of Sparse Defects in Coherent Imaging

2026

Second author, IEEE SPL (under review) | forward-model simulation, full evaluation harness

- The sample is a known flat background with a few unknown defects. The paper states how many scan positions recover the defects. The answer is two. One scan is provably not enough. A single scan cannot tell a defect from its mirror image. This project changed how I approach evaluation. I now ask first whether the measurement can determine the answer. I compare methods after that.
- I did data curation and software. I wrote the forward-model simulation. I wrote the whole evaluation harness with Gaussian probes, scan geometries, random defect sets, detector noise. I compared the proposed two-scan decoder against full-data linearization, against the phase-retrieval baselines PIE and HT-WF. I swept defect separation. I swept signal-to-noise ratio.

Open-Set Standpoint Screening with LLM Agent Panels

2026

Fifth author, EMNLP 2026 Main Conference | evaluation set, experimental analysis

- I built the evaluation set. I ran part of the experimental analysis. The system, MARS-Gov, drops the usual assumption that every bias type is known in advance. A Scout agent proposes affected groups that no fixed list names. Fixed and dynamic Juror agents debate over retrieved legal and policy text. A Judge agent verifies the outcome. A last agent rewrites the text. The meaning stays the same. The system reached an F1 score of 0.880, 20.2 points above the best zero-shot LLM. A leave-one-category-out test showed that truly unseen categories are still found, not memorized.

Controlled Electrochemistry | undergraduate thesis, Guangxi University

2025 – 2026

Sole author, advisor Prof. Meinan Liu | experimental design, three measurement channels, all analysis

- I ran a single-variable study on lithium-metal anode interfaces. I fixed every quantity except one process parameter. I carried an untreated baseline through the whole study as a parallel control. No direct image of the interface exists. Every claim had to come from agreement between independent measurements.
- I used three cell types as three measurement channels, the half cell, the symmetric cell, the full cell. A claim that held in one channel only could not be reported as a general improvement. The plating plateau overpotential fell from 145 to 114 mV. The nucleation dip fell from about 30 to about 20 mV. Symmetric cells stayed near 30 mV for 330 h. Full cells kept 85.3% capacity after 600 cycles.
- I reported the failures on both sides of the best setting. Too little treatment leaves bare sites. The cell fails at cycle 45. Too much treatment builds a layer that cracks. The cell fails at cycle 30. I measured the mechanism in two more channels, impedance spectroscopy and X-ray diffraction. The performance claim rests on an identified cause, not on a correlation. I kept the whole route on commercial reagents and standard hardware. Other labs can repeat it.

Services Founded and Run

Kangjie Returning-Student Tutoring Service, Yuncheng, Shanxi | Founder

Winter 2026

- I founded and run a non-profit tutoring service. The tutors are university students at home for the holiday. I handle scheduling, tutor–student matching, marketing, daily operations.
- I am rebuilding the workflow around AI agent tools. The tools now do coordination work that people did by hand. This can cut the staff cost of the service by a large margin. It tests whether such agents hold up in daily operation, not only in a demo. The service placed 28 tutors with 37 high-school students.

On-Demand Fitness Coaching Service, Guangxi University | Co-Founder

2026 – present

- I co-founded a live two-sided service with the Guangxi University Rehabilitation Association. It matches certified student coaches to users for one-on-one sessions on site. I own operating strategy, coach supply, the business model.

Industry Experience

Sixteen manufacturing firms in six industries, Guangxi | four supervised placements

2024 – 2026

- I followed the aluminium chain end to end. The chain runs from bauxite mining, alumina refining, electrolysis, casting, extrusion, rolling, battery foil. I visited steel plants, car body plants, construction machinery plants, solar module lines, superhard material plants. I stood at the inspection points where plant data is produced. This shapes how I judge a deployed model. A plant rates a system by throughput, yield, ease of inspection. One headline number counts for little.
- **Closest to my own work.** At **Nanning Industrial Investment Aluminium-Based New Materials Group** I followed the full route from ingot to foil for 12-micrometre aluminium foil. This foil is the cathode current collector in a lithium battery. The plant makes it by the hundred thousand tonne. I hand-assemble the same part into coin cells.

Leadership, Honors, and Technical Skills

Leadership. Captain, Guangxi University Football Team | President, University Football Association | President, Student Union, School of Resources, Environment and Materials.

Honors. Silver Award, Guangxi University Career Planning Competition (2023) | Third Prize, Guangxi University Metallography Contest (2023) | First Place (2025), Third Place (2024), Guangxi “Qianli Cup” Intercollegiate Football League, undergraduate division.

Programming and tools. Python, PyTorch, L^AT_EX, MATLAB, C, Origin | **Machine learning.** Supervised fine-tuning of open-weight LLMs, large-scale batched inference, LLM-as-judge pipelines, retrieval-augmented generation, multi-agent systems, checkpoint selection, output filtering | **Evaluation.** Factorial design, parallel controls, replication arms, holdout sets, coverage checks, failure analysis, stated limits | **Signal and imaging.** Forward-model simulation, phase retrieval (PIE, HT-WF), sparse recovery, sample complexity | **Instruments.** Electrochemical workstations, EIS, XRD, SEM, XPS, glovebox